SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 18

TITLE : Write a Java GUI program to demonstrate button click event handling.

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

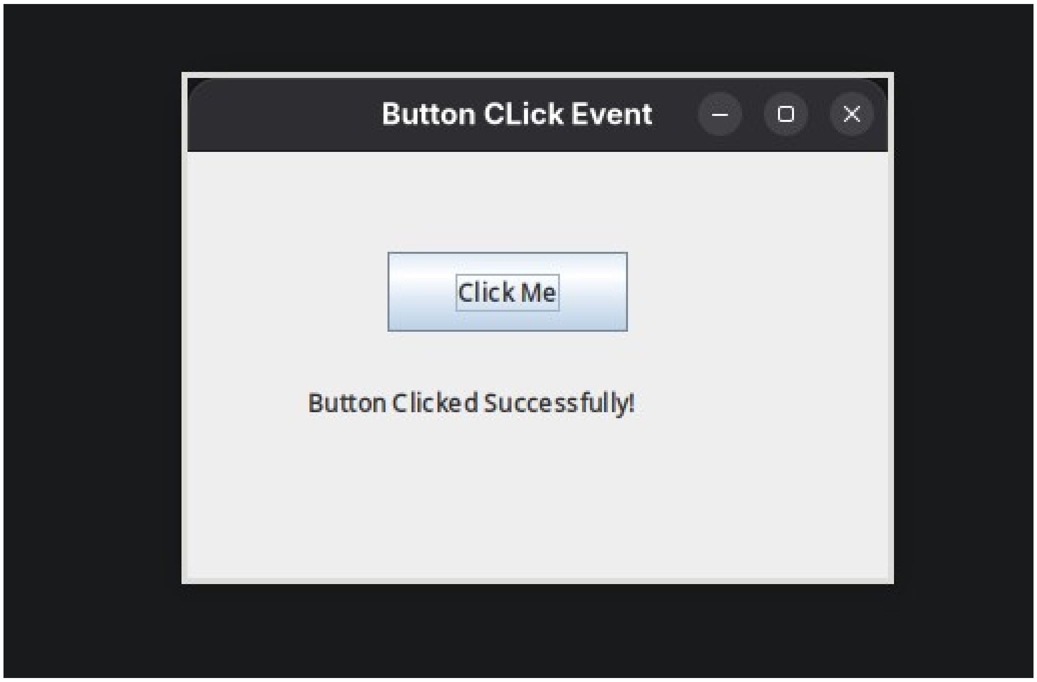
Learning Objectives

To implement button click events using Java event handling mechanisms.

Theory

Event handling allows Java programs to respond to user actions such as button clicks, mouse movements, and keyboard input. Swing uses ActionListener to process button click events. Event-driven programming separates user interface design from event-processing logic, making applications interactive and user-friendly.

Output Screenshots



CODE 1

```
package Exp18;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.*;

public class Main {
   public static void main (String[] args) {
         JFrame frame = new JFrame("Button CLick Event");
         frame.setSize(350, 250);
         frame.setLayout(null);
         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
               JButton button = new JButton("Click Me");
               button.setBounds(100, 50, 120, 40);

               JLabel label = new JLabel();
               label.setBounds(60, 110, 250, 30);

```

```
            button.addActionListener(new ActionListener() {
                    public void actionPerformed(ActionEvent e) {
                        label.setText("Button Clicked Successfully!");
                    }
            });

        frame.add(button);
            frame.add(label);
            frame.setVisible(true);
    }
}
```

EXERCISE 1

```
package Exp18.Exercise;

import javax.swing.*;

public class Exp1 {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Calculator");

        JTextField num1 = new JTextField();
        JTextField num2 = new JTextField();

        JButton add = new JButton("Add");
        JButton subtract = new JButton("Subtract");

        num1.setBounds(50, 40, 200, 30);
        num2.setBounds(50, 80, 200, 30);

        add.setBounds(50, 130, 100, 30);
        subtract.setBounds(160, 130, 100, 30);

        frame.add(num1);
        frame.add(num2);
        frame.add(add);
        frame.add(subtract);
```

```java

        frame.setSize(320, 220);
        frame.setLayout(null);
        frame.setVisible(true);

        add.addActionListener(e -> {
            int a = Integer.parseInt(num1.getText());
            int b = Integer.parseInt(num2.getText());
            JOptionPane.showMessageDialog(frame, "Result: " + (a+b));
        });

        subtract.addActionListener(e -> {
            int a = Integer.parseInt(num1.getText());
            int b = Integer.parseInt(num2.getText());
            JOptionPane.showMessageDialog(frame, "Result: " + (a-b));
        });
    }
}
```

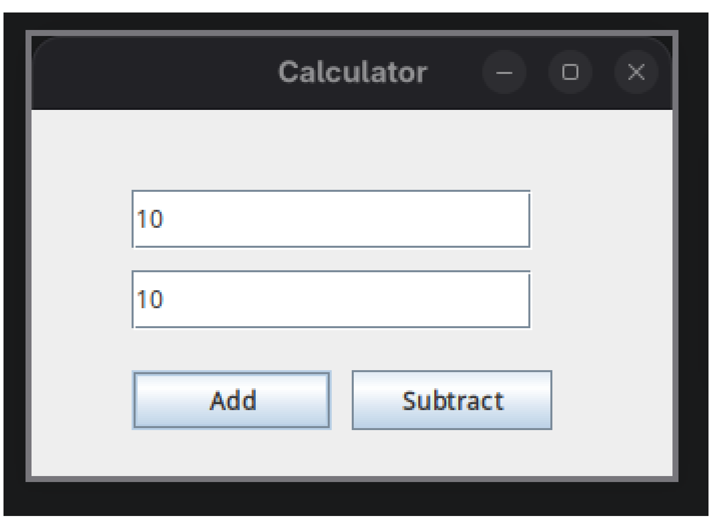
EXERCISE 2

```
package Exp18.Exercise;

import javax.swing.*;

public class Exp2 {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Bank Balance Calculator");

        JLabel balanceLabel = new JLabel("Initial Balance:");
        JLabel amountLabel = new JLabel("Transaction Amount:");

        JTextField balanceField = new JTextField();
        JTextField amountField = new JTextField();

        JButton deposit = new JButton("Deposit");
        JButton withdraw = new JButton("Withdraw");

        balanceLabel.setBounds(30, 40, 130, 30);
        balanceField.setBounds(160, 40, 180, 30);

        amountLabel.setBounds(30, 90, 130, 30);
        amountField.setBounds(160, 90, 180, 30);

        deposit.setBounds(70, 150, 100, 30);
        withdraw.setBounds(200, 150, 100, 30);

        frame.add(balanceLabel);
        frame.add(balanceField);
        frame.add(amountLabel);
        frame.add(amountField);
        frame.add(deposit);
        frame.add(withdraw);

        frame.setSize(400, 250);
        frame.setLayout(null);
        frame.setVisible(true);

        deposit.addActionListener(e -> {
```

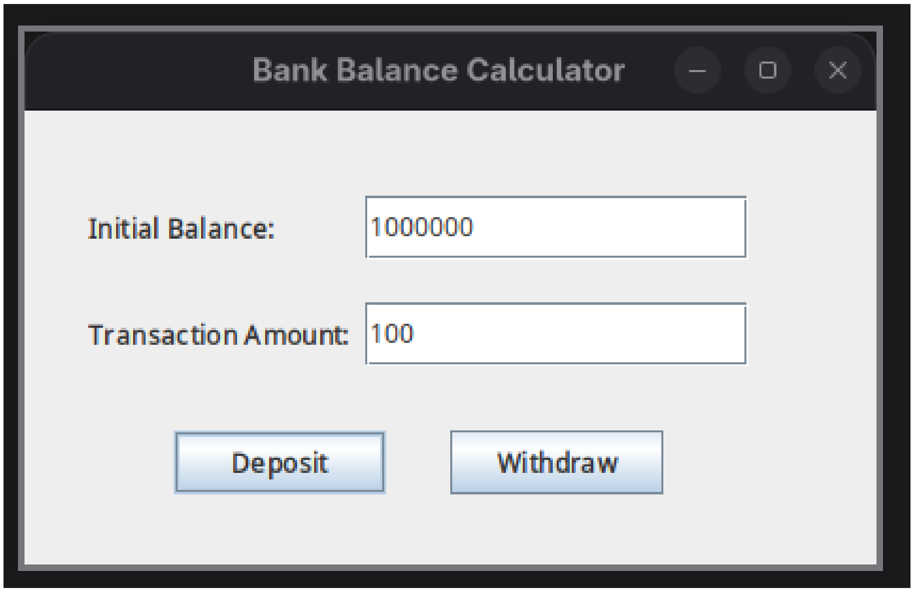
```java
          double balance=Double.parseDouble(balanceField.getText());
          double amount=Double.parseDouble(amountField.getText());

            balance = balance + amount;

            JOptionPane.showMessageDialog(
                frame,
                "Updated Balance: " + balance
            );
        });

        withdraw.addActionListener(e -> {
          double balance=Double.parseDouble(balanceField.getText());
          double amount=Double.parseDouble(amountField.getText());

            balance = balance - amount;

            JOptionPane.showMessageDialog(
                frame,
                "Updated Balance: " + balance
            );
        });
    }
}
```

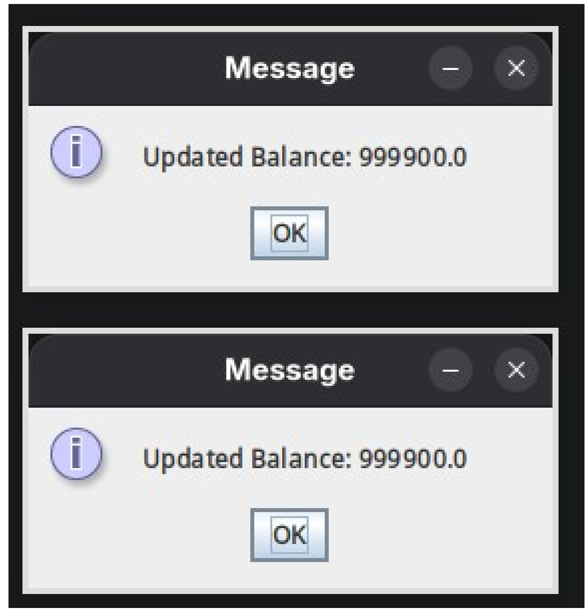
Exercise

Q1. Create a GUI calculator where buttons perform addition and subtraction.



Q2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.





Conclusion

This experiment helped in understanding the use of Java Swing GUI components for developing interactive applications. The registration form was created using components such as `JFrame`, `JLabel`, `JTextField`, `JButton`, and `JOptionPane`. The experiment demonstrated how user input can be collected, processed through event handling, and displayed effectively using a dialog box.